

Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm.

1. **F2** PUT THE YELLOW CORNERS HOME  
Yellow edges only – ignore the corners

2. **F2'** NOT TO WORRY edges at all

3. **F2L** FRONT FACE 2 TURNS LEFT  
Hook two at a right angle; hold them BLACK and LEFT

4. **F2R** FRONT FACE 2 TURNS RIGHT  
Line hold the two left-to-right

5. **F2L'** FRONT FACE 2 TURNS LEFT OPPOSITE  
Front face 2 turns up to three turns; up to hook to move to center

6. **F2R'** FRONT FACE 2 TURNS RIGHT OPPOSITE  
The algorithm is called 'strange' you use it on every card after this

7. **F2L'** FRONT FACE 2 TURNS LEFT OPPOSITE  
Two yellow faces RIGHT

8. **F2R'** FRONT FACE 2 TURNS RIGHT OPPOSITE  
Two yellow faces RIGHT

9. **F2L'** FRONT FACE 2 TURNS LEFT OPPOSITE  
Two yellow faces RIGHT

10. **F2R'** FRONT FACE 2 TURNS RIGHT OPPOSITE  
Two yellow faces RIGHT

11. **F2L'** FRONT FACE 2 TURNS LEFT OPPOSITE  
Two yellow faces RIGHT

12. **F2R'** FRONT FACE 2 TURNS RIGHT OPPOSITE  
Two yellow faces RIGHT

13. **F2L'** FRONT FACE 2 TURNS LEFT OPPOSITE  
Two yellow faces RIGHT

14. **F2R'** FRONT FACE 2 TURNS RIGHT OPPOSITE  
Two yellow faces RIGHT

15. **F2L'** FRONT FACE 2 TURNS LEFT OPPOSITE  
Two yellow faces RIGHT

16. **F2R'** FRONT FACE 2 TURNS RIGHT OPPOSITE  
Two yellow faces RIGHT

### FIRST SOLVE

Now here? Do the walk-through once at cubepath3.svercel.app/turn. This card is the memory jog.

**NOTATION**  
**R U F L B B** = turn that face clockwise, looking at from outside the cube  
**r** = counter-clockwise, **2** = half turn. Lowercase **r f** = that face plus the layer behind it, turning together.  
**y** = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

**WORDS**  
 algorithm = a fixed sequence of turns - trigger = a short algorithm you hand out to use **unit**

### 1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No YELLOW - work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

### 2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



**white sticker RIGHT**  
**R'U'R**



**white sticker FRONT**  
**L'U'L**



**white sticker UP** run the first one once to knock it out, then look again  
**R'U'R**

### 3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

**goes RIGHT**  
**U R'U' y L'U'L**



**goes LEFT**  
**U L'U'L U' y' R'U'R**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

R **R'U'R** sexy move   
 R **U'U** Sune   
 R **F'F'** sledhammer   
 gap = change grip

[illegible]

### FIRST SOLVE

New? Do the walk-through once on cubepath3.svercel.app/learn. This one's the memory jog.

**NOTATION**  
**R** = turn the face clockwise, looking at it from outside the cube.  
**r** = counter-clockwise, 2 = half turn. Lowercase **r f** = that face plus the layer behind it, turning together.  
**y** = spin the whole cube. Arrow shows twisting down. A whole-cube turn solves nothing.

**WORDS**  
 algorithm = a fixed sequence of turns • trigger = a short algorithm you can use to start a unit

#### 1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out. Can't see it? Build the four white edges on the YELLOW face – first, above its matching center, then turn each one down 180 degrees.

### 2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



**white sticker RIGHT**  
**RUR'U'**



**white sticker FRONT**  
**L'ULU**



**white sticker UP** run the first one once to knock it out, then look again  
**RUR'U'**

#### 3 TOP EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. If the arrow shows which slot it goes to.

**goes RIGHT**  
**U RURU' y L'ULU**



**goes LEFT**  
**U' L'ULU y' RUR'U'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ **RUR'** sexy move   
 ■ **RURU** Sune   
 ■ **RFRF** sledgehammer   
 gap = change grip

Diagram illustrating the sequence of operations for a 2D convolution, showing the input volume and the resulting output volume.

The input volume is a 3x3x3 cube, with the top layer (yellow) and bottom layer (blue) highlighted. The output volume is a 3x3x2 cube, with the top layer (yellow) and bottom layer (blue) highlighted.

The operations are labeled as follows:

- 6 PUT THE YELLOW CORNERS HOME (ignore the corners)
- 5 FURROW FUR (right angle)
- 4 FURROW FUR (right angle)
- 3 FURROW FUR (right angle)
- 2 FURROW FUR (right angle)
- 1 FURROW FUR (right angle)
- 2 TWIST THE YELLOW CORNERS (twist)
- 3 FURROW FUR (right angle)
- 4 FURROW FUR (right angle)
- 5 FURROW FUR (right angle)
- 6 PUT THE YELLOW CORNERS HOME (ignore the corners)

The diagram also includes a 3D coordinate system and a 2D grid of 3x3x3.

### FIRST SOLVE

New here? Do the walk-through once at [cubepath3.svevel.app/learn](#). This card is the memory jog.

#### NOTATION

**R U F L D B** = turn that face clockwise, looking at it from outside  
**r u f l d b** = counter-clockwise, 2 = half turn. Lowercase **r f** = that face plus the cube's top layer, turning together.  
**y spin** the whole cube left, white staying down. A whole-cube turn solves nothing.

#### WORDS

algorithm = a fixed sequence of turns • trigger = a short algorithm you handsolve as one unit

#### 1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it by hand.

Can't see it? Build the four white edges on the YELLOW face first, each about its matching center, then turn each one down 180 degrees.

#### 2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.

|                                                                                     |                                             |
|-------------------------------------------------------------------------------------|---------------------------------------------|
|  | <b>white sticker RIGHT</b><br><b>'R'U'</b>  |
|  | <b>white sticker FRONT</b><br><b>'L'L'U</b> |
|  | <b>white sticker UP</b><br><b>'R'U'R'</b>   |

#### 3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

**goes RIGHT**  
**U' R'U' y L'L'U**

**goes LEFT**  
**U' L'L'U' y' R'U'R'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

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**R'UR'** sexy move    **R'UR'** Sune    **R'FRF'** sled/changer gap + change grip

[illegible]

### FIRST SOLVE

New here? Do the walkthrough once at [cubepath-53.vercel.app/learn](http://cubepath-53.vercel.app/learn). This is the memory log.

**NOTATION**  
**R** = turn that face clockwise, looking at it from outside the cube  
**r** = counter-clockwise, 2 = half turn. Lowercase **r f** = that face plus the layer behind it, turning together  
**y** = spin the whole cube left. Whole turning. A whole-cube turn solves nothing.

**WORDS**  
 algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run on one unit

### 1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

### 2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



**white sticker RIGHT**  
**RUR'U'**

**WORDS**  
 white sticker FRONT  
**L'ULU**

**white sticker U' run the first one once to knock it out, then look again**  
**RUR'U'**

### 3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

**goes RIGHT**  
**U' L'ULU y L'ULU**

**goes LEFT**  
**U' L'ULU y' L'ULU**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

**RUR'U'** sexy move   **RUR'U** Sune   **R'FRF** sledgehammer   gap = change grip